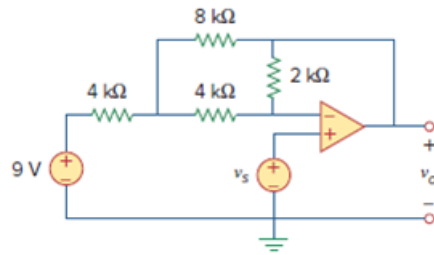


Problem

In the circuit of Fig. 5.59, calculate v_o if $v_s = 2$ V.

Figure 5. 59:



Step-by-step solution

Step 1 of 4

Refer to the circuit diagram in Figure 5.59 in the textbook.

The circuit is labeled as shown in Figure 1.

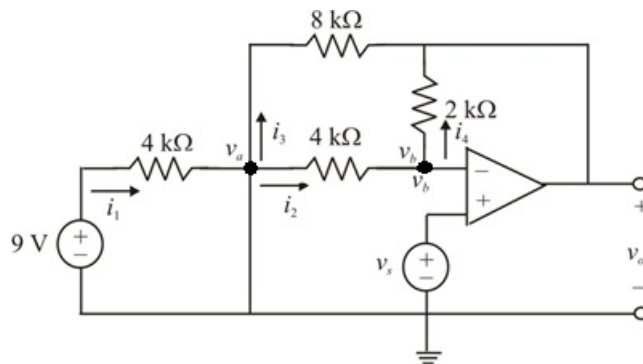


Figure 1

Step 2 of 4

The source voltage $v_s = 2\text{ V}$

Applying Kirchhoff's current law (KCL) to node 1.

$$i_1 = i_2 + i_3$$

$$\frac{9 - v_a}{4} = \frac{v_a - v_b}{4} + \frac{v_a - v_o}{8}$$

$$18 - 2v_a = 2v_a - 2v_b + v_a - v_o$$

$$5v_a - 2v_b - v_o = 18$$

Therefore $5v_a - 2v_b - v_o = 18$

But, $v_b = v_s$ since non-inverting terminal is connected to voltage source v_s , above equation becomes:

$$5v_a - 2v_s - v_o = 18$$

Substituting the value of $v_s = 2\text{ V}$ in the equation.

$$5v_a - 2v_s - v_o = 18$$

$$5v_a - 2(2) - v_o = 18$$

$$5v_a - v_o = 22$$

Therefore $5v_a - v_o = 22 \dots\dots (1)$

Step 3 of 4

Applying Kirchhoff's current law (KCL) to node 2,

$$i_2 = i_4$$

$$\frac{v_a - v_b}{4} = \frac{v_b - v_o}{2}$$

$$v_a - v_b = 2v_b - 2v_o$$

$$v_a - 3v_b + 2v_o = 0$$

But, $v_b = v_s$ since non-inverting terminal is connected to voltage source v_s , above equation becomes:

$$v_a - 3v_s + 2v_o = 0$$

Step 4 of 4

Substituting the value of $v_s = 2\text{ V}$ in the equation.

$$v_a - 3v_s + 2v_o = 0$$

$$v_a - 3(2) + 2v_o = 0$$

$$v_a + 2v_o = 6 \dots\dots (2)$$

Multiply equation (2) by 5.

$$5v_a + 10v_o = 30 \dots\dots (3)$$

Solve equation (1) and (3).

Subtract equation (1) from equation (3).

$$5v_a + 10v_o = 30$$

$$5v_a - v_o = 22$$

$$11v_o = 8$$

$$v_o = 727.27\text{ mV}$$

Therefore, the output voltage v_o is 727.27 mV .